

Universal Approximation Theorem (UAT)

Cybenko 1989

σ -activation, continuous

$$\sigma: \mathbb{R} \rightarrow \mathbb{R} \quad \begin{cases} \lim_{x \rightarrow -\infty} \sigma(x) = 0 \\ \lim_{x \rightarrow +\infty} \sigma(x) = 1 \end{cases}$$

Fix error rate $\epsilon > 0$:

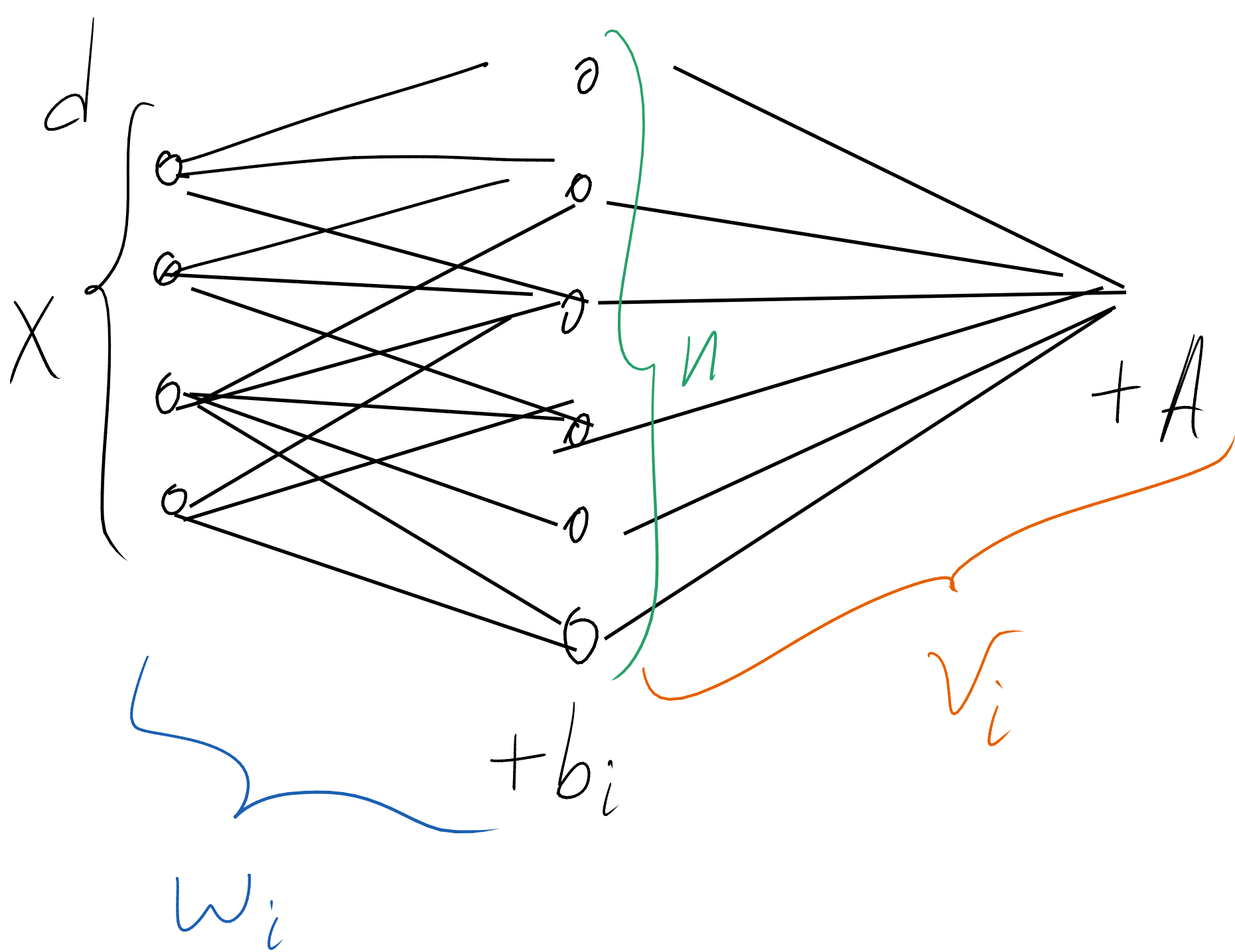
$$\forall f \in C([0, 1]^d) \quad f: \mathbb{R}^d \rightarrow \mathbb{R}$$

$$\exists n \in \mathbb{N} \quad \{w_i\}_{i=1}^n \quad w_i \in \mathbb{R}^d$$

$$\{b_i\}_{i=1}^n \quad b_i \in \mathbb{R}$$

$$\{v_i\}_{i=1}^n \quad v_i \in \mathbb{R} \quad A \in \mathbb{R}$$

$$g(x) = \sum_{i=1}^n v_i \sigma(w_i^T x + b_i) + A$$



$$\sup_{x \in [0,1]^d} |f(x) - g(x)| \leq \varepsilon$$

Optimization

$$\mathcal{L}^t(w) = \frac{1}{B} \sum_{i=1}^B \mathcal{L}(f_w(x_{t_i}), y_{t_i})$$

$t_i \sim U(\{1, \dots, \ell\})$

$$g_t = \nabla_w \mathcal{L}^t(w_{t-1})$$

SGD

$$w_t = w_{t-1} - \eta_t g_t$$

$$\begin{cases} g_t = \nabla_w \mathcal{L}^t(w_{t-1}) \\ m_t = \mu m_{t-1} + g_t \\ w_t = w_{t-1} - \eta_t m_t \end{cases}$$

SGD +
momentum

$$m_0 = 0 \quad 0 \leq \mu < 1 \quad \mu = 0.9$$

$$\mathcal{L}^t(w) \rightarrow \mathcal{L}_\lambda^t(w) = \mathcal{L}^t(w) + \frac{\lambda}{2} \|w\|_2^2$$

$$\nabla_w \|w\|_2^2 = 2w$$

$$\begin{cases} g_t = \nabla_w \mathcal{L}^t(w_{t-1}) + \lambda w_{t-1} \\ m_t = \mu m_{t-1} + g_t \\ w_t = w_{t-1} - \eta_t m_t = \end{cases}$$

$$\begin{aligned}
&= w_{t-1} - \eta_t (\mu m_{t-1} + g_t) = w_{t-1} - \eta_t (\mu m_{t-1} + \\
&+ \nabla_w \mathcal{L}^t(w_{t-1}) + \lambda w_{t-1}) = \\
&= \underbrace{w_{t-1}(1 - \lambda \eta_t)}_{\text{weight decay}} - \underbrace{\eta_t (\mu m_{t-1} + \nabla_w \mathcal{L}^t(w_{t-1}))}_{\approx m_t^{\text{old}}}
\end{aligned}$$

$$\lambda \sim 10^{-4}, 10^{-5}$$

Adam

RMS prop + Momentum



AdaGrad

$$\left\{ \begin{aligned}
g_t &= \nabla_w \mathcal{L}^t(w_{t-1}) + \lambda w_{t-1} \\
m_t &= \beta_1 m_{t-1} + (1 - \beta_1) g_t \\
v_t &= \beta_2 v_{t-1} + (1 - \beta_2) g_t \circ g_t \quad \leftarrow \text{element-wise product} \\
\hat{m}_t &= \frac{m_t}{1 - \beta_1^t} \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t} \\
w_t &= w_{t-1} - \eta_t \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}
\end{aligned} \right.$$

$m_0 = 0$
 $v_0 = 1$

$\epsilon, \beta_1, \beta_2$ — hyperparameters

m_t — momentum

v_t — adaptive learning rate

$$\epsilon \approx 10^{-8}, \quad \beta_1 = 0.9, \quad \beta_2 = 0.999$$

$$m_t = (1-\beta_1)g_t + \beta_1 m_{t-1} = (1-\beta_1)g_t + \beta_1(1-\beta_1)g_{t-1} + \beta_1^2 m_{t-2} = \sum_{i=0}^{t-1} (1-\beta_1) \beta_1^i g_{t-i}$$

$$\hat{m}_t = \frac{1}{1-\beta_1^t} \cdot m_t = \frac{(1-\beta_1)}{1-\beta_1^t} \sum_{i=0}^{t-1} \beta_1^i g_{t-i} =$$

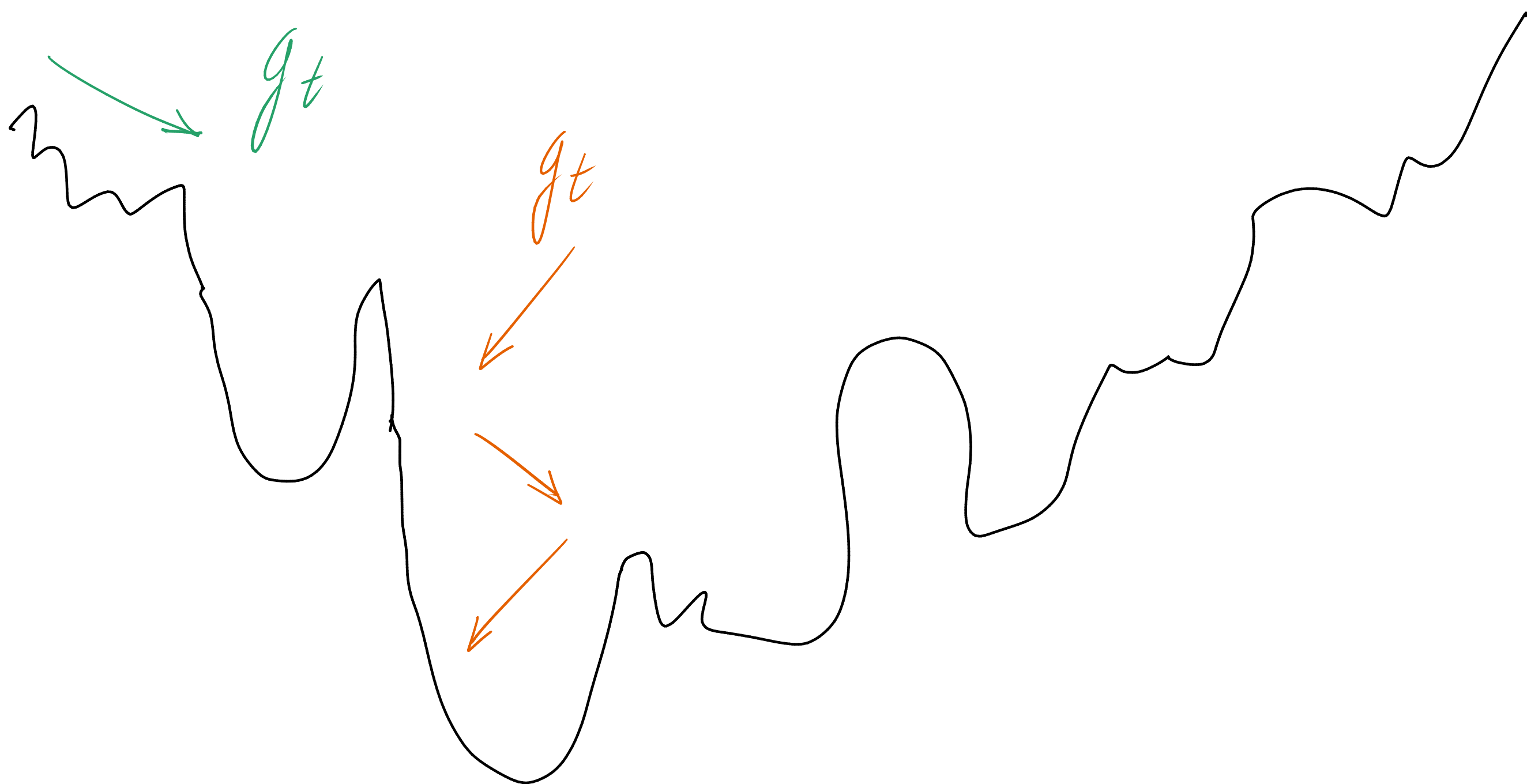
$$= \frac{\sum_{i=0}^{t-1} \beta_1^i g_{t-i}}{\sum_{i=0}^{t-1} \beta_1^i} \quad (\equiv)$$

$$\alpha_i = \frac{\beta_1^i}{\sum_{j=0}^{t-1} \beta_1^j}$$

$$\sum_{i=0}^{t-1} \alpha_i = 1$$

$$\quad (\equiv) \quad \sum_{i=0}^{t-1} \alpha_i g_{t-i} = \mathbb{E} g$$

$$\hat{v}_t = \mathbb{E} g \odot g = \text{Var}(g) + \mathbb{E} g \odot \mathbb{E} g$$



$$\text{Var}(g) \approx 0$$

$$w_t = w_{t-1} - \eta_t \frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}} \approx$$

$$\approx w_{t-1} - \vec{\eta}_t$$

Far from optimum

$$\frac{\mathbb{E} g}{\sqrt{\cancel{\text{Var}(g)} + \mathbb{E} g \odot \mathbb{E} g + \epsilon}} \approx 1$$

0 ←

Near optimum

$$\text{Var}(g) \gg 0 \quad \frac{\mathbb{E}g}{\sqrt{\text{Var}(g) + \mathbb{E}g \odot \mathbb{E}g + \epsilon}} < 1$$

SGD + momentum

Adam

convolutional

RNN, Transformer

$$w_t = w_{t-1} - \eta_t \frac{\hat{w}_t}{\sqrt{\hat{v}_t + \epsilon}}$$

$$= w_{t-1} - \eta_t \frac{m_t}{(1 - \beta_1^t) \sqrt{\hat{v}_t + \epsilon}}$$

$$= w_{t-1} - \eta_t \frac{\beta_1 m_{t-1} + \nabla_w \mathcal{L}^t(w_{t-1}) + \lambda w_{t-1}}{(1 - \beta_1^t) \sqrt{\hat{v}_t + \epsilon}}$$

$$= w_{t-1} \left(1 - \frac{\eta_t \lambda}{(1 - \beta_1^t) \sqrt{V_t + \epsilon}} \right)$$

AdamW

$$\left\{ \begin{array}{l} g_t = \nabla_w \mathcal{L}^t(w_{t-1}) + \cancel{\lambda w_{t-1}} \\ m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t \\ v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t \odot g_t \\ \hat{m}_t = \frac{m_t}{1 - \beta_1^t} \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t} \\ w_t = \cancel{w_{t-1}} - \eta_t \frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}} \end{array} \right.$$

$$w_{t-1} (1 - \lambda \eta_t)$$

SAM - Sharpness-aware minimization

η_t - ? learning rate

- Const scheduler

$$\eta_t = \eta_0$$

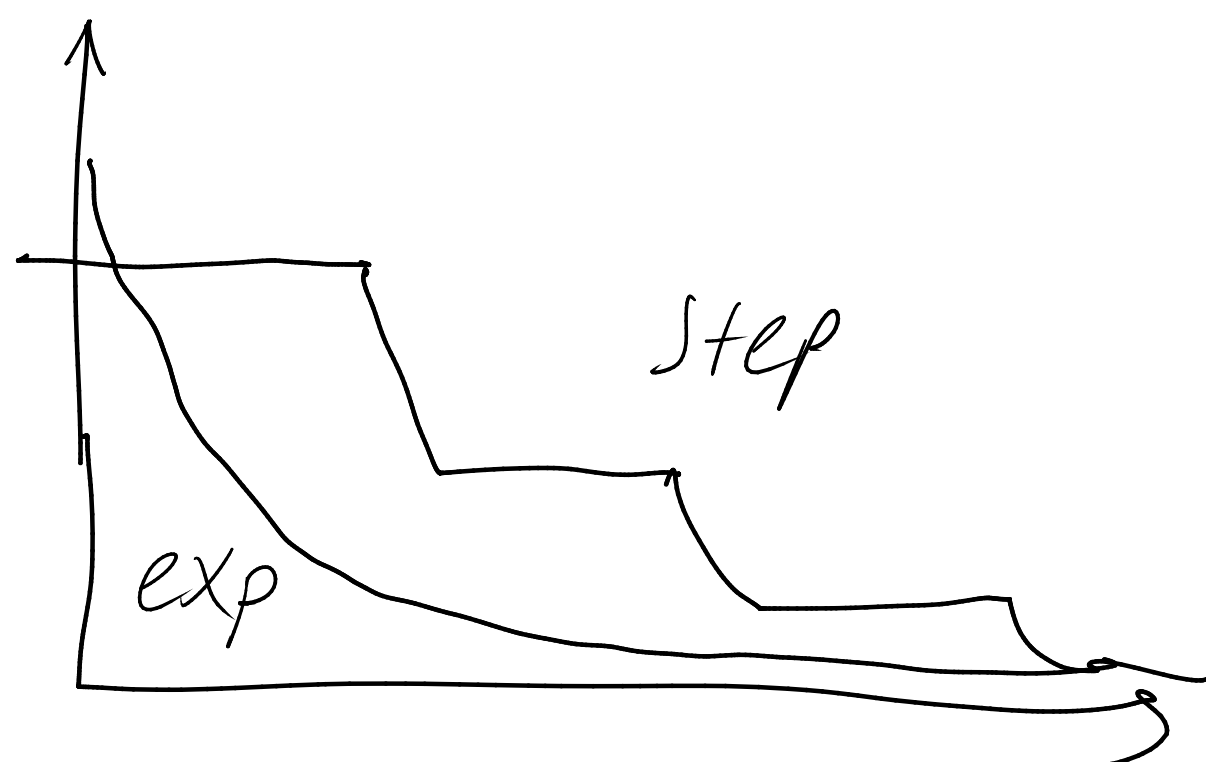
- MultiStepLR

$$\eta_t = \begin{cases} \eta_0 & 0 < t \leq t_1 \\ \eta_1 & t_0 < t \leq t_2 \\ \vdots & \end{cases}$$
$$\eta_0 > \eta_1 > \dots$$

- ExponentialLR

$$\eta_t = \gamma \cdot \eta_{t-1}$$

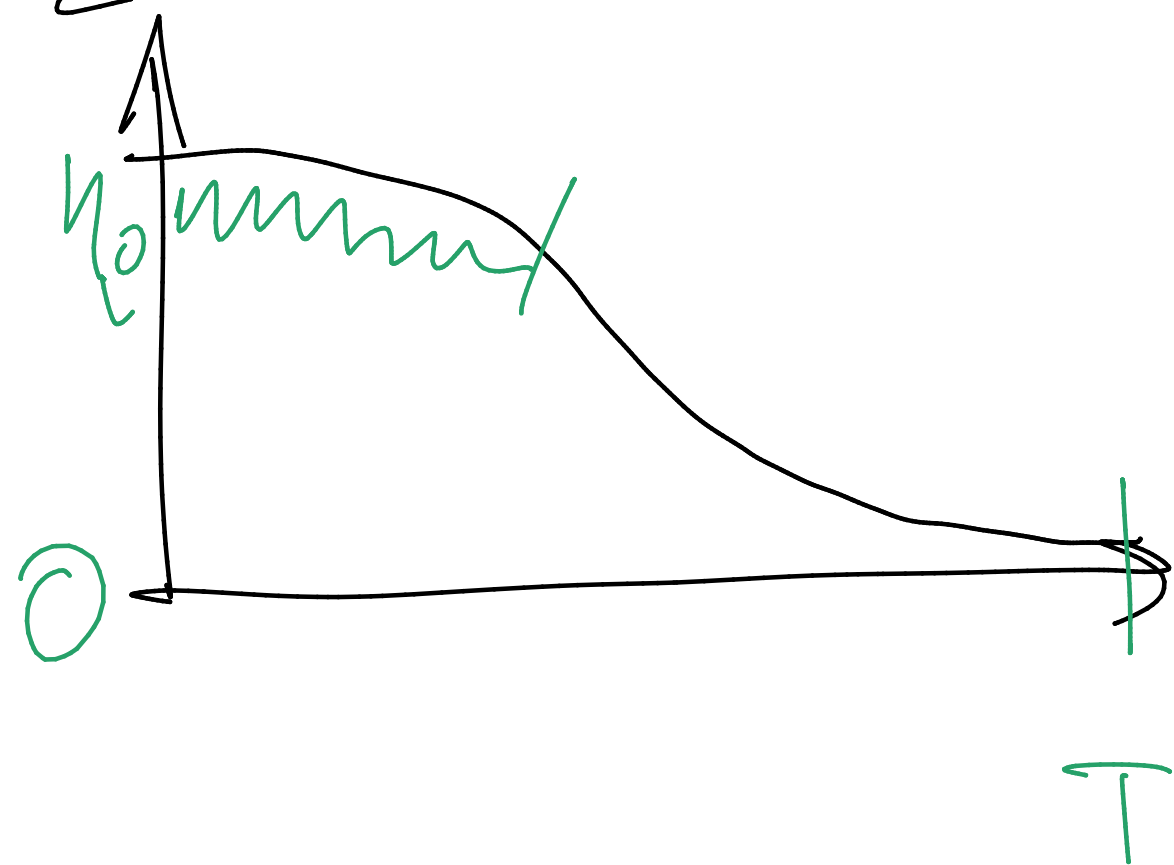
$$\gamma < 1$$



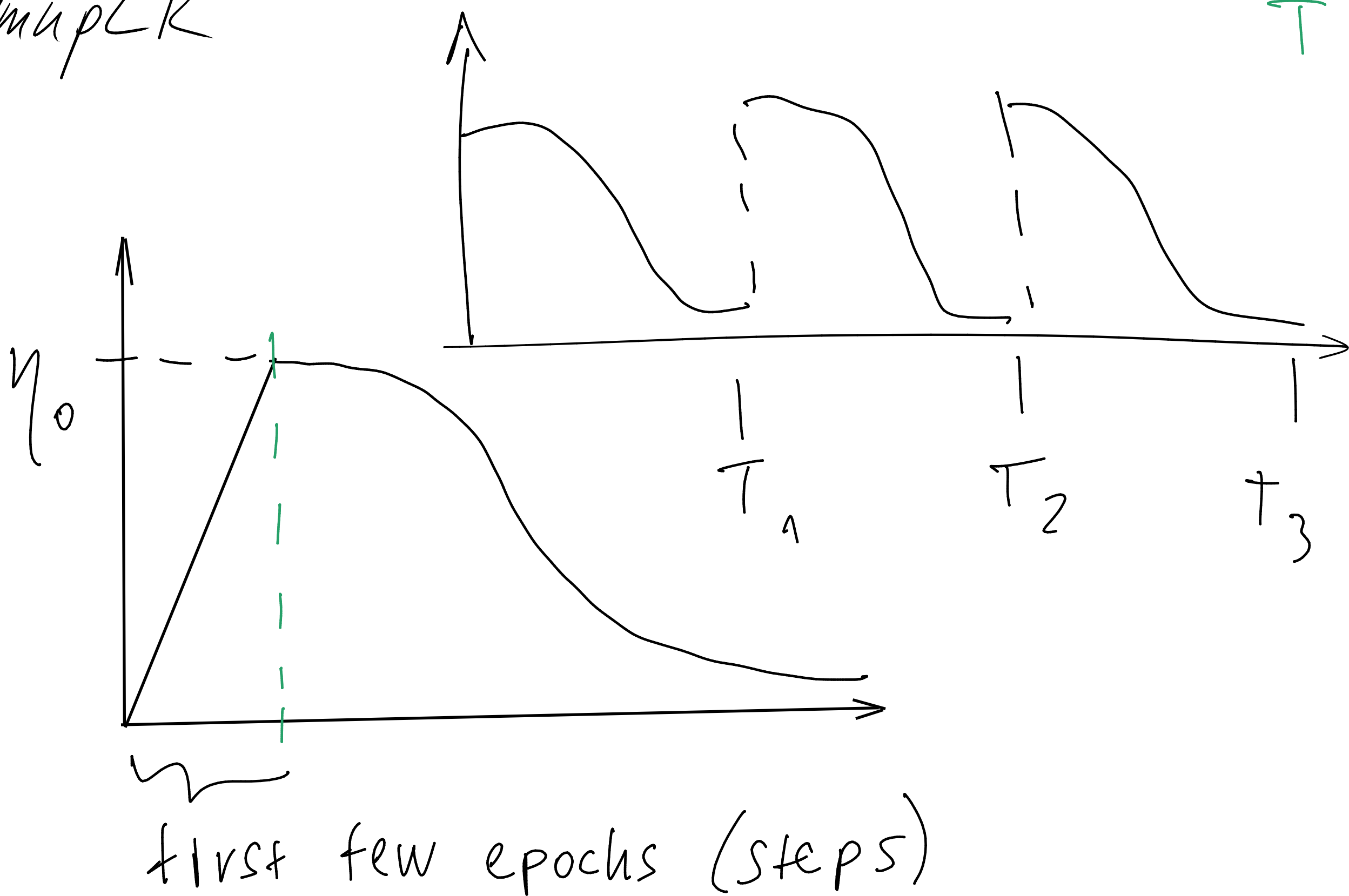
- Cosine LR

$$\eta_t = \eta_0 \cdot \frac{1}{2} \left(1 + \cos \left(\frac{\pi t}{T} \right) \right)$$

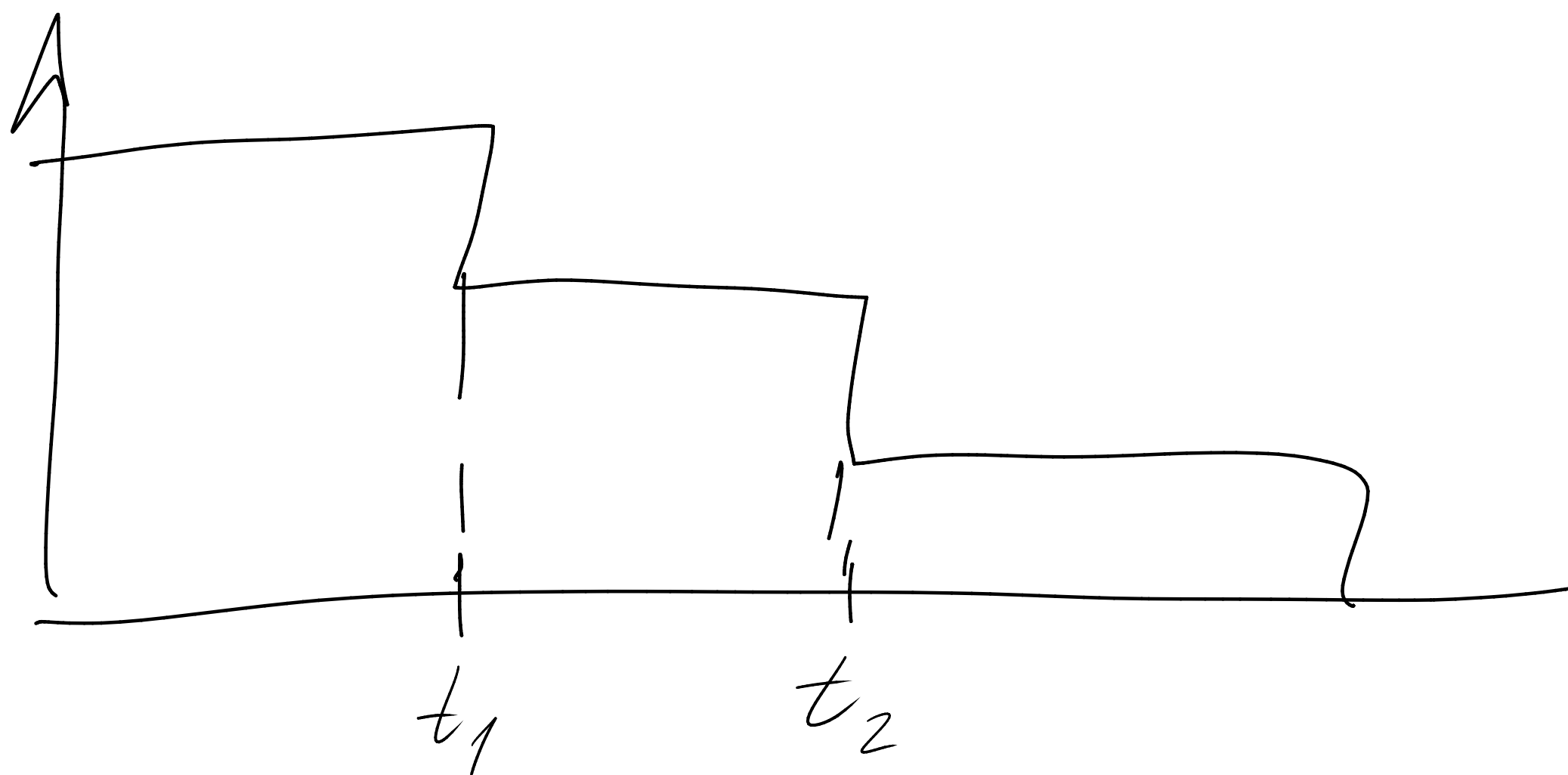
T - total number of steps



- Warmup LR



- Reduce LR On Plateau



t_1, t_2 - on validation metric